

ADAM MESSAOUDI

Data Engineer | Business Intelligence Engineer | Data Analyst

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PROFESSIONAL SUMMARY

Data Engineer experienced in ETL/ELT pipelines, data modeling, Microsoft Fabric Lakehouse, Power BI dashboards, SQL, Python automation, AI-assisted development, and business intelligence solutions. Skilled in transforming complex datasets into actionable insights and building scalable analytics platforms. Currently refining German language skills (B1 level) and motivated to contribute to international data engineering environments.

PROFESSIONAL EXPERIENCE

End-of-Studies Data Engineering Intern 03/2025 – 09/2025
ZF Group Hanover, Germany

Project: Enterprise Data Platform & Predictive Analytics for Global Workshop Networks

- Architected and deployed a scalable enterprise data platform using Microsoft Fabric (OneLake, Dataflows Gen2, Semantic Models) and Salesforce, integrating 130,000+ records into a centralized Lakehouse environment through automated ETL/ELT pipelines.
- Automated data extraction from 4,600+ global workshops using Python and Selenium, enriching CRM data with external Google ratings and review metrics through integrated data workflows.
- Developed 5 interactive Power BI dashboards with semantic models and a Workshop Health Score (0–100), combining audit compliance, customer reputation, and operational KPIs for performance monitoring.
- Delivered advanced analytics solutions including an NLP-based Data Assistant using spaCy/OpenAI API and an ML forecasting pipeline using MLflow and regression models to predict workshop network trends.
- Technologies:** Microsoft Fabric (OneLake, Dataflows Gen2), Power BI (Power Query, DAX), Python (Pandas, Selenium, spaCy, Scikit-Learn), OpenAI API, MLflow, Flask, REST APIs, SQL, Salesforce.

Data Engineering Intern 06/2024 – 09/2024
Infraway Tunisia Tunis, Tunisia

Project: Telecom Management Optimization Through Data-Driven Insights

- Analyzed 10.01M+ telecommunications call records across 43 attributes via EDA to uncover key network traffic trends.
- Streamlined ETL workflows by building automated Talend staging jobs, reducing data processing time by 45% and enabling daily dashboard refreshes.
- Designed 2 Power BI dashboards (Service Quality & Profitability) for Sales Managers and Service Operators, optimizing 63 B2B accounts.
- Trained Exponential Smoothing forecasting models in Python (Pandas, Statsmodels) to predict monthly call volumes and support capacity planning.
- Technologies:** MySQL, Talend, Power BI, Python, Machine Learning (Statsmodels, Scikit-Learn), SQL.

Business Intelligence Intern 06/2023 – 09/2023
PGS International Ben Arous, Tunisia

Project: Data-Driven Inventory Optimization Platform for Oil Depots

- Centralized oil depot inventory data across 5 depot locations into a PostgreSQL data warehouse using automated Talend ETL workflows.
- Visualized real-time stock levels and 10+ operational KPIs across 3 Power BI dashboards to track inventory performance.
- Implemented Row-Level Security (RLS) and optimized semantic models using Power BI and Tabular Editor.
- Orchestrated Python web scrapers (Selenium, BeautifulSoup) parsing market data across 4 target industry sources daily to enrich analytics.
- Optimized 10+ backend REST API routes of a Flask analytics platform to support data-driven inventory management.
- Technologies:** Talend, PostgreSQL, Power BI (Row-Level Security (RLS), Tabular Editor), Python, Angular, Flask, Machine Learning, Web Scraping (Selenium, BeautifulSoup).

TECHNICAL SKILLS

- Data Engineering & BI:** Microsoft Fabric (OneLake, Lakehouse, Dataflows, Pipelines), Power BI (DAX, Power Query, RLS), ETL/ELT, Data Modeling, Data Warehousing.
- Programming & Databases:** Python (Pandas, Flask, Selenium), SQL (PostgreSQL, MySQL, Oracle, MS SQL Server), MongoDB, REST APIs, Automation.
- AI & Machine Learning:** Machine Learning, Predictive Modeling, Time Series Forecasting, Scikit-Learn, MLflow, NLP, OpenAI API, Azure OpenAI, Generative AI.
- Cloud, Enterprise & Tools:** Azure (Storage, Azure SQL, Entra ID), Salesforce, Dynamics 365, SAP BW, Git, Docker, Linux, Agile/Scrum.

CERTIFICATIONS

SAP Solution Implementation – SAP | Coursera 11/2024
Microsoft Azure Cloud Services – Microsoft | Coursera 10/2024
MLOps (Machine Learning Operations) – Duke University | Coursera 10/2024
Talend Data Integration – Talend | Coursera 10/2024

EDUCATION

Engineering Degree in Computer Science (Master's Equivalent) Graduated: 10/2025
ESPRIT School of Engineering – Tunisia

Exchange Program in Computer Science 10/2024 – 09/2025
Schmalkalden University of Applied Sciences – Germany

Baccalaureate in Experimental Sciences 08/2020
Borj Cedria High School – Tunisia

LANGUAGES

German: B1 (In Progress) | **English:** Fluent | **French:** Fluent | **Spanish:** Elementary | **Arabic:** Native